Akoizo Original Practice - Form 3 - Answer key

Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Micas are representative…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Sheet silicates

B. Native elements

C. Halides

D. Sulfides

Correct: A

2. (2 points) Angular gravel clasts suggest…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Breccia

B. Conglomerate specifically

C. Obsidian

D. Schist

Correct: A

3. (2 points) A rock can contain…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Only one chemical element

B. No geological material

C. Only transparent crystals

D. Multiple minerals

Correct: D

4. (2 points) The approved way to distinguish halite is…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Taste every sample

B. Heat an unknown

C. Assume all clear crystals are salt

D. Physical evidence without tasting

Correct: D

5. (2 points) The original rock is the…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Streak

B. Ore grade

C. Unit cell

D. Protolith

Correct: D

6. (2 points) Acicular describes…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. A hardness value

B. A chemical reaction

C. Needlelike habit

D. One unique species

Correct: C

7. (2 points) Bowen’s series is…

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. A hardness scale

B. A river classification

C. A conditional crystallization model

D. Every rock’s mandatory sequence

Correct: C

8. (2 points) Under uncertainty, a useful answer states…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Only a color

B. An unrelated process

C. Evidence and a plausible alternative

D. Unjustified certainty

Correct: C

9. (4 points) Compare pyroxene and amphibole cleavage.

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

answer: Pyroxene commonly approaches 90°; amphibole commonly approaches 56°/124°. (4 points).

10. (4 points) Differentiate erosion and deposition.

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

answer: Erosion removes material; deposition accumulates it. (4 points).

11. (4 points) Distinguish face from cleavage.

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

answer: A growth face forms during crystallization; cleavage is a preferred breakage direction. (4 points).

12. (4 points) Name a carbonate and oxide.

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

answer: Calcite is a carbonate; hematite is an oxide. (4 points).

13. (4 points) Why are index minerals conditional?

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

answer: Stability depends on pressure, temperature, composition, and reaction history. (4 points).

14. (4 points) How can veins constrain timing?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: A vein that cuts host material is generally younger than the material it cuts. (4 points).

15. (4 points) Contrast glass and fine crystals.

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Glass lacks long-range crystalline order; fine-grained rock contains small crystals. (4 points).

16. (4 points) What is a poor timed strategy?

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

answer: Browsing many unlabeled images without narrowing candidates by evidence. (4 points).

17. (4 points) Why can quartz and feldspar differ?

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

answer: They differ in chemistry, substitutions, and detailed lattice bonding despite both being frameworks. (4 points).

18. (4 points) Why avoid a unique environment from one grain size?

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

answer: Different agents and flow histories can produce overlapping grain populations. (4 points).